MHS: Modular Plug-and-Play (M-PNP) Workstream Overview, Updates, and Progress Towards Industry Adoption

Dave Heinrich *Fellow, Compute Business Unit, HPE*

Chitrak Gupta *Senior Director Product Management, AMI*

Phil Leech *Principal Technologist / Architect, OCP M-PNP Workstream Lead, HPE*

Todd Rosedahl *Director, Systems and AI Strategy Jabil*

AMI, HPE

Jabil

Panel Discussion



Dave Heinrich

Fellow
Compute Business Unit, HPE



Chitrak Gupta

Senior Director Product Management, AMI



Phil Leech

Principal Technologist / Architect
OCP MHS M-PNP Workstream Lead
Compute Business Unit, HPE



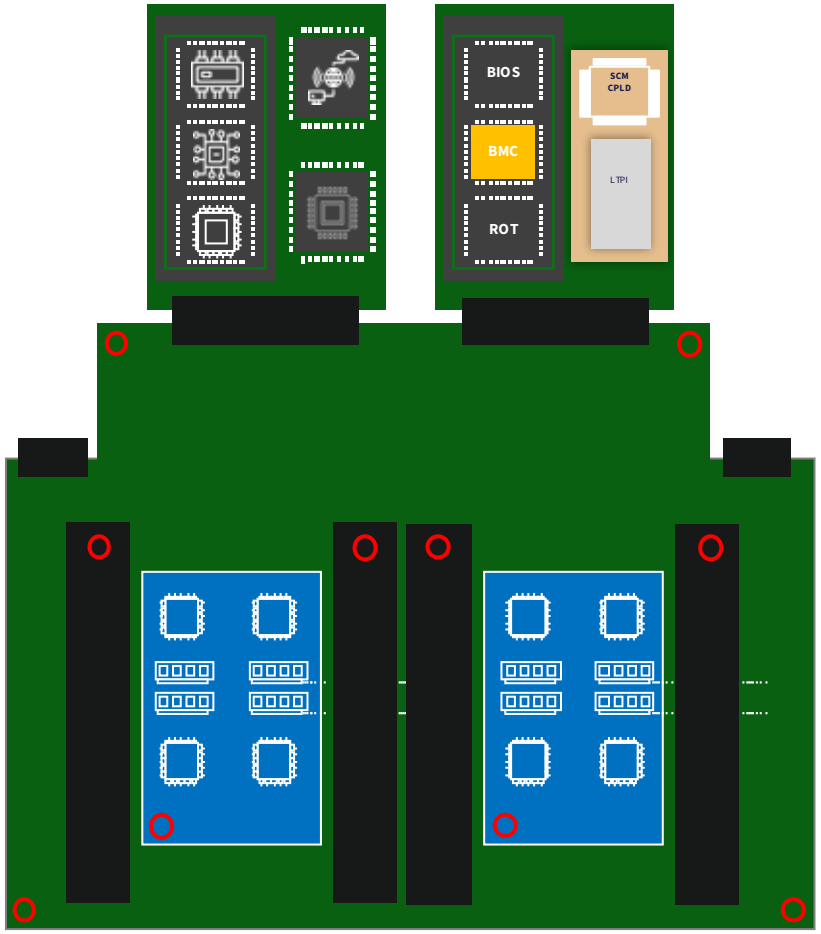
Todd Rosedahl

Director
Systems and AI Strategy Jabil

Agenda

- Motivation and Plug-and-Play Vision
- M-PNP Workstream and Specifications Overview
- Progress, Plans and Challenges
- Schedule & Call to Action

Plug-and-Play Overview – What and Why



What:

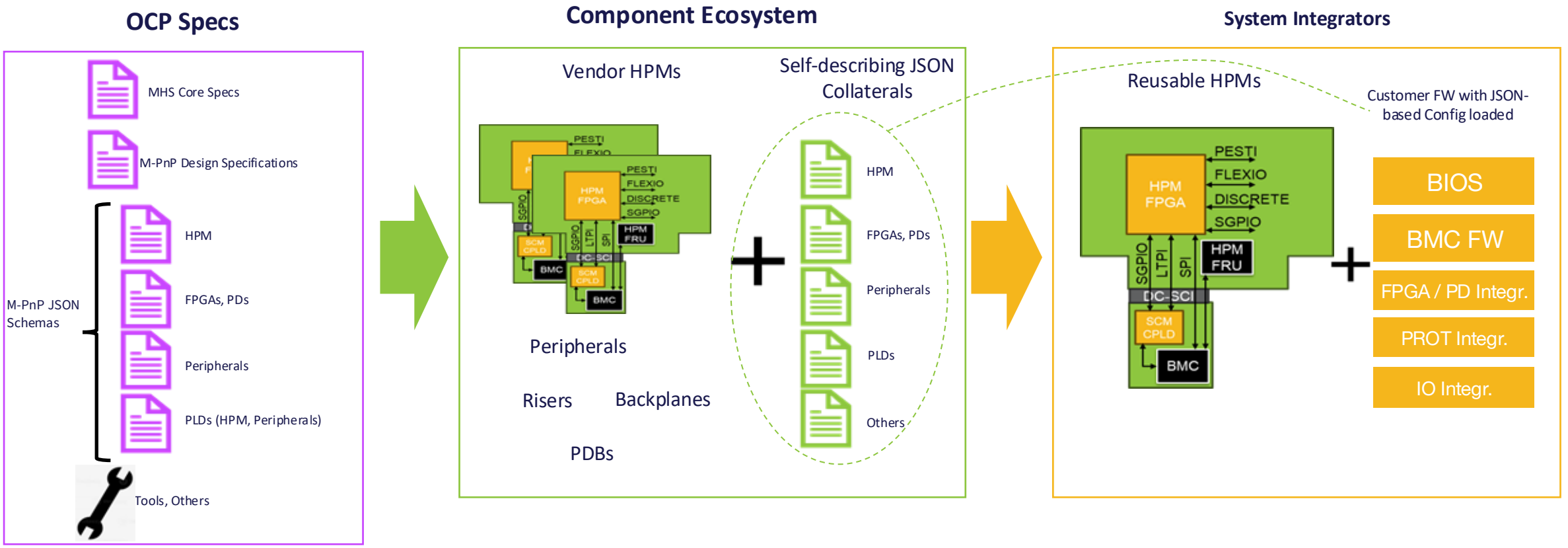
- Current specifications allow HW (SCM/HPM) interoperability
  - FW changes are required for proper system functionality (Plug-and-Code)
- Enable component interoperability (SCM/HPM) with no code changes.
  - Ease-of-interoperability is foundational to modular HW success.
- Define self-describing HW definition files to enable Plug-and-Play
- Commoditize and automate non-value-added platform enablement functions

Why:

- Time To Market for diverse system capabilities
- Lower OpEx & CapEx to integrate and validate
- Generational re-use of HW with cost & sustainability benefits
- Ease customer specific value-add and security solutions
- Enables alternate HW sourcing strategies, reuse, & leverage
- Standard framework and tools are conducive to System Workflow Automation

The Modular Plug and Play (M-PNP) Vision

M-PNP Vision: Fully Re-usable Hardware with Baseline Plug-and-Play



The MPNP Journey: From Plug-and-Code to Plug-and-Play

We Are Here

Interoperability Framework | **Interoperability Realization**

Plug & Code

Must pre-build all code with HPM/Chassis/System Knowledge

Plug & Play 'Lite'

- Some design specs defined
- Can compile in or dynamically load some JSON Files
- Must compile in other definition files

Plug & Play

- Runtime Loading of all definition files
- All knowledge is provided to the BMC for all HW components
- Provides baseline functionality

Design Specification Work:

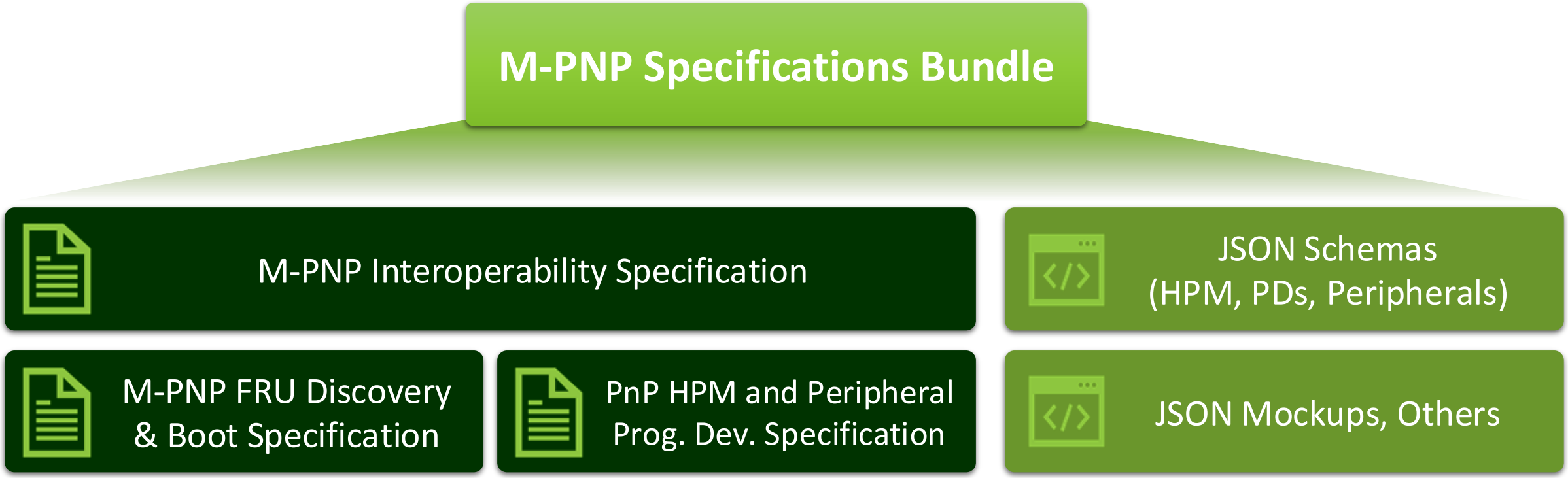
- Interoperability Specification(s)
- Supporting Design Specifications (e.g., component FRU, data models, APIs, etc.)

Further Design & Implementation Specification Work:

- Interoperability specifications completed
  - OEM extensions incorporated
- Conformance Requirements defined
- Additional Supporting Design Specifications

Workstream Deliverables

- Main specs delivered as the M-PNP Specifications Bundle (at 0.7)
- Contains additional collaterals such as tools, schemas, mockups, etc.

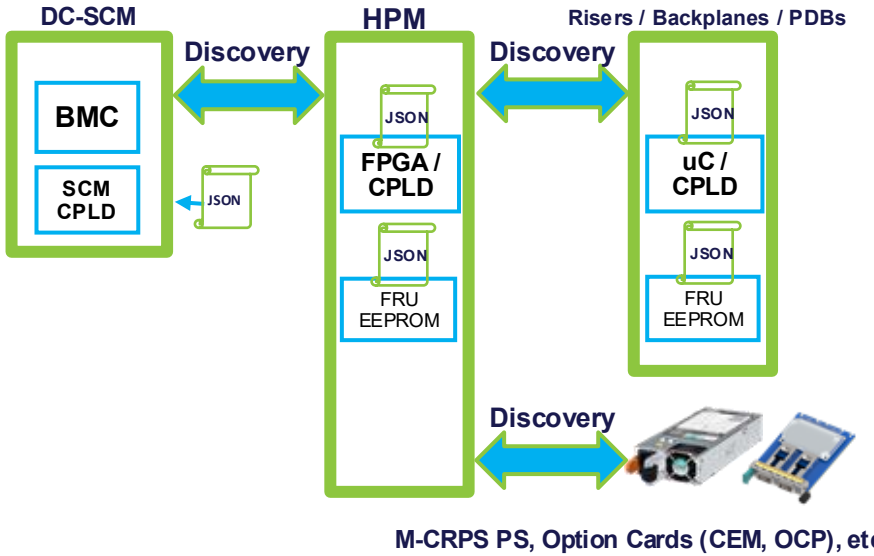


M-PNP Specification Suite

Three specifications delivered in the M-PNP Specifications Bundle (v0.7)

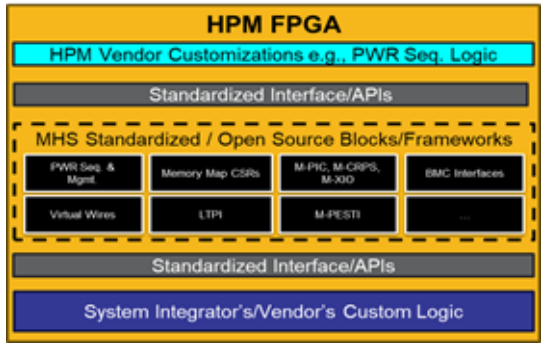
M-PNP.FRU_DISCOVERY_BOOT

- Component discovery by SCM BMC firmware
- Unique FRU information per component
- Self-describing JSON schema for FRU data
- System boot requirements for plug and play
- Spans DC-SCM, HPM, risers/backplanes/PDBs



M-PNP.HPM_PERIPHERAL_PD (Was FPGA)

- Standardized HPM and Peripheral Programmable Device (PD) interfaces for SCM BMC firmware (buses, protocols) (LTPI, SPI, I2C/I3C, SGPIO)
- Framework for mapping IP blocks, memory map, virtual wires
- JSON schema for IP blocks, CSRs, virtual wires
- Security, versioning and other non-functional requirements



HPM FPGA: standardized interfaces over MHS open-source blocks

M-PNP.INTEROP

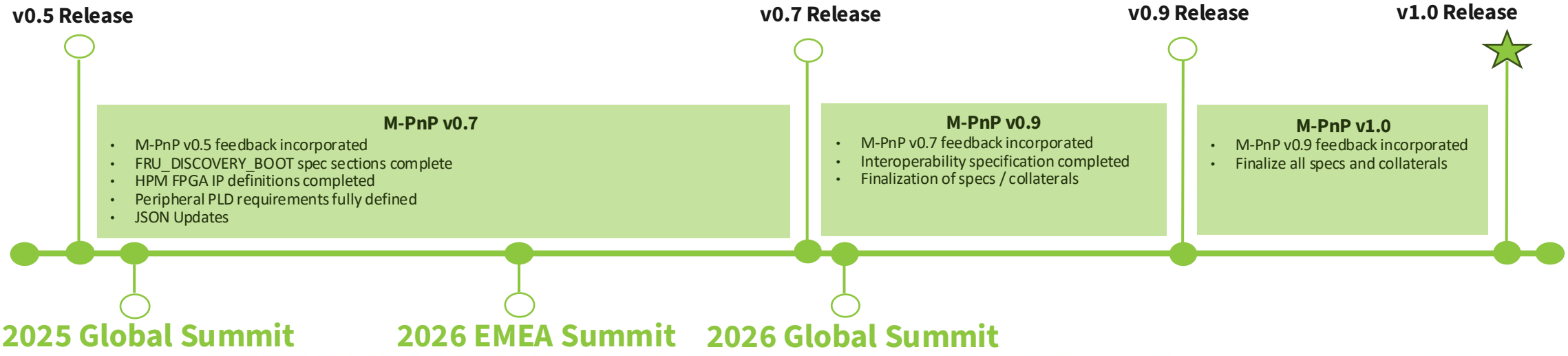
- Scope, M-PNP vision and definition of interoperability
- Interoperability requirements and conformance
- Ties the two design specs to the wider spec set

Supporting design specifications

- OCP DC-MHS, DC-SCM 2.1 & HPM Type 1 common circuits
- Existing peripheral specs (e.g., OCP NIC)
- Automatable HW design map templates
- Common security specs (e.g., SPDM), other industry specs (DMTF, PCIe)

M-PNP Workstream Progress and Schedule

- **M-PnP v0.5 Specs released prior to OCP Global Summit ’25**
  - M-PNP.FRU_DISCOVERY_BOOT, M-PNP.FPGA, and M-PNP.INTEROP specifications
  - HPM and Peripheral FPGA / Programmable Device JSON schemas and example mockup files
  - FRU Packaging Tool (frupack.py) was contributed to OCP by HPE
- PD JSON file creation tool (fpga2json.py) to contribute to OCP by HPE targeted Q4 ‘26
- **M-PnP v0.7 Specs released Oct ’26 – Soliciting Feedback now via webinars, outreach, etc.**
- Prototype JSON files, FPGA IP blocks, and packaging/retrieval of FRU information files Q3 / Q4 ‘26
- **M-PnP v0.9 targeted after OCP Global Summit Q4 ‘26 with 1.0 in Q1 ‘27**



What's New In The 0.7 Spec Package?

- M-PNP.FRU_Discovery_Boot Specification 0.7
  - Peripheral guidance, discovery process error paths, security, updated JSON, and more
- M-PNP.HPM_Peripheral_Programmable_Device Specification 0.7
  - Workstream voted to change the name to broaden the scope of the specification to cover peripheral programmable devices as well as HPM FPGAs
  - Feature complete IP blocks, M-PESTI virtual wire definitions, updated JSON, etc.
- M-PNP.INTEROP Interoperability Specification 0.3
  - Working on update to 0.7 outside of package release or at 0.9 package
- Updated JSON Schemas and Mockups
  - Schema documentation added
- Tools and Examples (deliver targeted in Q4 '26 / Q1 '27)

M-PNP Workstream Tools, Contributions, and POC Efforts

- **Prototyping Tools**
  - AMI Contributions: AI driven platform FRU JSON generation Tool
  - Lattice Contributions: FPGA JSON Tool, PnP Reference Designs, EEPROM integrations
  - HPE Contributions (Current and Planned): FRU Packaging Tool, PD JSON Builder (fpga2json.py), Net2JSON (net2json.py), JSON2DeviceTree (json2devtree.py)
  - Intel Contributions: JSON Checker Tool
- **Contributions**
  - Lattice FPGA IP block Verilog reference code
  - Joint HPE-Intel-Lattice POC white paper
- **POCs**
  - MPNP Innovation Station: HPE-Intel-Lattice joint POC, Lattice Dual Node POC
  - AMI Booth: M-PNP in Action

Challenges & Final Thoughts

- Platform / chassis level self-descriptive JSON
  - Codifying platform behaviors such as LED strategy and chassis thermal requirements
  - How to create, load, and store this information
  - Extended information necessary to holistically manage set of components (e.g., PCIe adapters)
- Prototyping fully M-PNP supporting platform
  - Creating JSON files, implemented standardized FPGA IP blocks, prototyping BMC code, providing reference code, BIOS support, etc.
  - Integration of all firmware and programmable logic: HPM FPGA, Peripheral Programmable Devices, BMC and BIOS
- Collecting industry feedback on specifications content
  - We need your help by reviewing specs and Tech Talks content
  - Future Tech Talks possible (get industry feedback)

Call To Action

Help us get to Plug-and-Play!

- Review the Tech Talks and collateral covering M-PNP specifications topics:
  - M-PNP Overview and FRU_Discovery_Boot Tech Talk: https://www.youtube.com/watch?v=QapIKDsCq3E
  - M-PNP FPGA Tech Talk: https://www.youtube.com/watch?v=sWQxhUJzc3k
  - Specs: https://www.opencompute.org/w/index.php?title=Server/MHS/DC-MHS-Specs-and-Designs#Latest_DC-MHS_Specifications
- Provide feedback through DC-MHS Project or get directly involved in the WS
  - Bring-in ideas & contribute to the design & interoperability specifications
- Help us prototype – use the open-source tools to create your own files (coming soon)
  - FRU Packaging Tool: https://github.com/opencomputeproject/OCP-SVR-MHS-M-PNP_FRUTool
- Stop by the M-PNP Innovation Station and CLA member booths for demos / questions

Reach out to us through:

- DC-MHS Project Wiki: https://www.opencompute.org/wiki/Server/DC-MHS
- DC-MHS Public Mailing List: ocp-mhs-public@OCP-All.groups.io

Thank You!

